

Installation Guide

The analysis was conducted using Stata 19 and R 4.3+. Stata `.do` files are plain-text scripts that contain sequences of Stata commands, and R `.R` scripts generate all figures from the coefficient tables exported by Stata. Each `.do` file in our shared data folder corresponds to a specific part of the analysis. Readers who wish to replicate our code need to have both Stata and R installed.

Stata Installation

The analysis was conducted using **Stata 19**, which is available for Windows, macOS, and Linux operating systems. **Operating System:** Compatible with Windows 10 or later, macOS 10.13 or later, and most modern Linux distributions. Stata is **not open-source software**, so a valid **license and activation key** are required to use it.

- **Programming Language:** Stata (version 19).
- **Hardware Requirements:** No non-standard hardware or high-performance computing resources are needed.
- **Obtaining a License:** When you purchase Stata—either directly from StataCorp, through a distributor, or via your institution—you will receive a license email. This email typically includes a PDF or text containing your serial number, code, and authorization key, which serve as your license and activation credentials. If your organization or university provides Stata, the license and activation key are often available through your IT department or software distribution portal.
- **Stata installation guide:** You can follow the guidance to install the Stata from the official website, <https://www.stata.com/install-guide/>
- **Typical Install Time:** Installation of Stata 19 typically takes less than 15 minutes on a modern machine.

Stata Packages Required for Conducting the Analysis

In addition to built-in Stata packages, we installed the following user-written Stata packages for our analysis:

- **eventstudyinteract** package
eventstudyinteract is a Stata package that implements the interaction weighted estimator for an event study. Sun and Abraham (2021) proposes this estimator as an alternative to the canonical two-way fixed effects regressions with relative time indicators.

See detailed introduction of this package from the developer website: <https://github.com/lsun20/EventStudyInteract>

To install the **eventstudyinteract** package, execute the following in Stata:

```
ssc install eventstudyinteract, replace
```

or

```
github install lsun20/eventstudyinteract
```

which should install the dependency packages. If not working, try manually install

```
ssc install avar
ssc install reghdfe
ssc install ftools
```

In Stata. Thereafter, type to view the help file.

```
help eventstudyinteract
```

- **estout** package

estout is a Stata package used to export regression results and coefficient tables to CSV files for R figure generation.

To install **estout** on your system, run command

```
ssc install estout, replace
```

in Stata. Thereafter, type to view the help file.

```
help esttab
```

R Installation

All figures in the paper are generated using **R** (version 4.3 or later). **R** is **free, open-source software** available for Windows, macOS, and Linux.

- **R installation guide:** Download and install R from the official website: <https://cran.r-project.org/>
- **RStudio (recommended):** We recommend installing RStudio as an integrated development environment (IDE) for running R scripts. Download from: <https://posit.co/download/rstudio-desktop/>
- **Typical Install Time:** Installation of R and RStudio typically takes less than 10 minutes on a modern machine.

R Packages Required for Generating Figures

The following R packages are required. To install all packages at once, run the following command in the R console:

```
install.packages(c("ggplot2", "dplyr", "readr", "patchwork",
                  "stringr", "sf", "tmap", "tmtools", "haven"))
```

A brief description of each package:

- **ggplot2**: Creates the event study plots using a grammar of graphics approach.
- **dplyr**: Data manipulation (filtering, summarizing coefficient tables).
- **readr**: Reads CSV coefficient tables exported by Stata.
- **patchwork**: Combines multiple ggplot panels into composite figures.
- **stringr**: String manipulation for parsing Stata output.
- **sf**: Handles spatial data (shapefiles) for map generation.
- **tmap**: Creates thematic maps of Appalachian counties.
- **tmaptools**: Auxiliary tools for tmap (bounding boxes, cropping).
- **haven**: Reads Stata `.dta` files directly into R (used for map data).

To verify that all packages are installed correctly, run:

```
library(ggplot2); library(dplyr); library(readr); library(patchwork)
library(stringr); library(sf); library(tmap); library(tmaptools)
library(haven)
cat("All packages loaded successfully.\n")
```

Note on sf package: The `sf` package requires system-level geospatial libraries (GDAL, GEOS, PROJ). On macOS, these are typically installed automatically via Homebrew when installing `sf`. On Linux, you may need to install them first (e.g., `sudo apt install libgdal-dev libgeos-dev libproj-dev` on Ubuntu). On Windows, they are bundled with the R package.